

Rajdeep Gill

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EDUCATION

University of Manitoba

Bachelor of Science in Computer Engineering

GPA: 4.45/4.5

Relevant Coursework: Data Structures and Algorithms, Linear Algebra, Stochastic Processes, Parallel Computing, Computer Architecture, Operating Systems, Software Engineering, Embedded Systems, Digital Signal Processing

Expected Graduation: May 2026

Winnipeg, MB

EXPERIENCE

Biosensors Research Lab | Undergraduate Researcher

May 2024 - Dec 2024

- Designed and trained a convolutional neural network using Python and PyTorch to predict cell viability from 10,000+ diffraction images, achieving 85% accuracy, demonstrating potential for non-invasive cell viability assessment
- Improved particle diffraction centroid detection success rate by 17% (from 75% to 92%) using computer vision techniques with Python and OpenCV, enhancing data quality for CNN training
- Developed an interactive data visualization tool in Python with Tkinter to process, filter, and display diffraction patterns and model predictions
- Applied unsupervised machine learning (Gaussian Mixture Model) with Python and Scikit-learn to classify cells based on impedance measurements, identifying distinct groups related to cell health

Biosensors Research Lab | Undergraduate Researcher

May 2023 - Aug 2023

- Applied dimensionality reduction (t-SNE, PCA) using Python and Scikit-learn to analyze and cluster datasets, revealing key correlations guiding future research
- Optimized data analysis pipeline by utilizing parallel processing and optimizing code efficiency, by over 125%
- Developed an automated oscilloscope testing platform with Python and PyVISA, streamlining data collection protocols to enhance measurement speed and ensure repeatability

PROJECTS

Financial Dashboard | React, TypeScript, HTML/CSS, Next.js, Hono.js, PostgreSQL

- Developed a full-stack financial tracking dashboard using React and Next.js for dynamic data visualization and expense/income management
- Implemented a RESTful API using Hono.js for efficient handling of user authentication, transactions, and data queries
- Utilized PostgreSQL for robust relational data storage and management of user financial records

Deaf Curling Accessibility Aid | C, ESP-32, Fusion360

- Engineered an embedded system using C on an ESP-32 to enable real-time communication between deaf curlers
- Designed a custom PCB to integrate the ESP-32 with LEDs and buttons, and a 3D-printed case using Fusion360

Distributed Image Processing | C++, OpenMPI

- Implemented a distributed image processing system using OpenMPI to parallelize filtering tasks across a cluster.
- Achieved a 3.78x speedup for large image filtering (e.g., Gaussian blur) compared to serial execution by distributing computational load across 4 nodes using OpenMPI.

PUBLICATIONS

B. Arzhang, E. Kovacs, J. Lee, **R. Gill**, E. Salimi, D. Thomson, and G. Bridges, "Machine Learning Classification of Microplastics by Integrating Optical and Dielectrophoresis Features," in *IEEE AP-S/URSI 2025*.

TECHNICAL SKILLS

Programming Languages: Python, C, C++, Java, TypeScript, JavaScript

Frameworks & Libraries: PyTorch, OpenCV, OpenMPI, Tkinter, React, Node, Express, Spring Boot, Flask

Tools: Git, Jenkins, Jira, AWS, Linux, Docker, Firebase, SQL, NoSQL, Excel, REST API

Concepts: Backend Development, Frontend Development, Fullstack Development, Software Engineering, Machine Learning, Networking, CI/CD, Microservices, Infrastructure Engineering, DevOps, Automation, Agile